

# A Complete Mathematical Tutorial on Human–AI Productivity Paradoxes

Skill, Effort, Unreliability, and AI Literacy

Teaching notes for Aouad, Lykouris, and Zhong (2026)

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Scope: main paper and mathematical supplements A–C; the convex-cost extension is omitted

## Abstract

These notes rebuild the paper’s mathematical argument from first principles. Every optimization step, stationary-distribution calculation, derivative decomposition, threshold comparison, and mode argument needed for the three main results is written out. The central message is that AI is locally productive at a fixed skill, yet can lower aggregate or expected productivity through endogenous effort: effort affects future skill, responds to unreliable assistance, and interacts with skill-dependent verification ability. The notes are designed to support a graduate lecture sequence and include diagnostic checks, worked special cases, proof roadmaps, and exercises with solutions. Appendix D of the paper, which extends the model to convex effort costs, is intentionally outside the scope requested for these notes.

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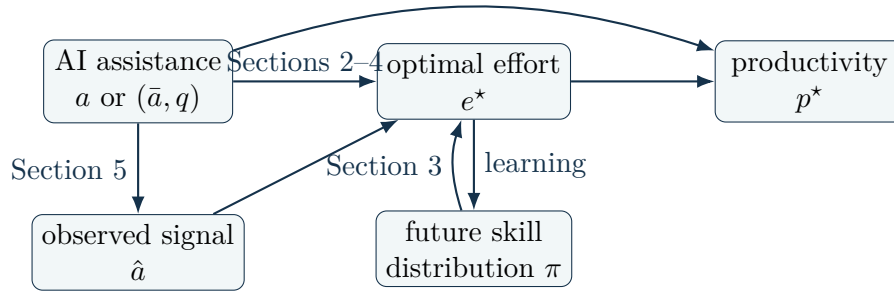
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# 1 How to read the paper

## 1.1 The three feedback loops

All of the paper’s results begin with the same static choice: an agent with skill  $s$  chooses costly effort  $e$  in the presence of AI assistance. The three main sections add one feedback loop at a time.



The logical order is therefore:

1. solve the deterministic static optimization exactly;
2. let effort control upward movement in a birth–death skill process;
3. make AI assistance random and differentiate through the endogenous optimizer;
4. let skill determine signal accuracy, then feed expected effort back into skill dynamics.

## 1.2 Main results and proof dependencies

| Result      | Economic statement                                                                                           | Mathematical engine                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic model | AI replaces effort and weakly raises productivity at fixed skill.                                            | Concavity of $p(x) - \gamma x$ and a change of variable from effort to total input.                                                                                        |
| Theorem 1   | With endogenous learning, productivity may rise and then fall within a fixed activity regime.                | Detailed balance; first-order stochastic dominance; a quotient derivative; a single-crossing result; large- $\mu$ asymptotics governed by the sensitivity gap $\Delta_m$ . |
| Theorem 2   | With unreliable AI, IARA gives a decreasing-then-increasing profile, while DARA gives an increasing profile. | Implicit differentiation of the effort FOC and factorization by the Arrow–Pratt coefficient $A(x) = -p''(x)/p'(x)$ .                                                       |
| Theorem 3   | Skill-dependent AI literacy can produce multiple modes in the long-run skill distribution.                   | Bayes’ rule; piecewise effort formulas; non-monotonicity in skill; the identity $\pi_{k+1}/\pi_k = \lambda(e_k)/\mu$ .                                                     |

## 1.3 Assumptions that do real work

The main analysis assumes:

1.  $s, e, a \geq 0$  enter production additively as  $x = s + e + a$ ;

2.  $p$  is nonnegative, increasing, continuous, twice differentiable where required, and concave;
3. effort cost is linear,  $c(e) = \gamma e$ ,  $\gamma > 0$ ;
4. agents are myopic: current effort does not internalize future skill;
5. in the dynamic model,  $\lambda(e) > 0$  is increasing (and, in Section 3, concave and differentiable);
6. skill decays at the constant rate  $\mu > 0$ .

The linear cost is important for the exact flat region of state-level productivity. The paper’s convex-cost extension is not treated here.

## 2 Mathematical toolkit

### 2.1 Concavity and the largest maximizer

Let  $g(x) = p(x) - \gamma x$ . Since  $p$  is concave,  $g$  is concave. The paper assumes

$$\limsup_{x \rightarrow \infty} \frac{p(x)}{x} < \gamma,$$

so  $g(x) \rightarrow -\infty$  at least along the tail strongly enough to make the argmax nonempty and bounded. Define the largest maximizer

$$x^* := \max \arg \max_{x \geq 0} \{p(x) - \gamma x\}.$$

For differentiable  $p$ , concavity gives the useful slope characterization

$$p'(x^*) \geq \gamma, \quad p'(x^* + \varepsilon) < \gamma \quad (\varepsilon > 0).$$

If  $p'(x) = \gamma$  over an interval, selecting the largest maximizer chooses its right endpoint. This tie-breaking rule is why all later formulas use  $x^*$  unambiguously.

### 2.2 Finite birth–death chains

For states  $1, \dots, N$ , let the upward rate from  $k$  to  $k + 1$  be  $b_k > 0$  and the downward rate from  $k + 1$  to  $k$  be  $d_{k+1} > 0$ . At stationarity, detailed balance is

$$\pi_k b_k = \pi_{k+1} d_{k+1}.$$

Hence, recursively,

$$\frac{\pi_{k+1}}{\pi_k} = \frac{b_k}{d_{k+1}}, \quad \pi_k = \pi_1 \prod_{i=1}^{k-1} \frac{b_i}{d_{i+1}}.$$

Normalization  $\sum_k \pi_k = 1$  yields

$$\pi_1 = \left( 1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{b_i}{d_{i+1}} \right)^{-1}.$$

This identity is used twice: first with  $b_k = \lambda(e^*(s_k, a))$ , then with  $b_k = \lambda(e_b^*(s_k, \bar{a}, q))$ .

## 2.3 First-order stochastic dominance on ordered states

For ordered skills  $s_1 \leq \dots \leq s_N$ , distribution  $X$  first-order stochastically dominates  $Y$  if

$$\sum_{i=1}^k \mathbb{P}(X = s_i) \leq \sum_{i=1}^k \mathbb{P}(Y = s_i), \quad k = 1, \dots, N.$$

The inequality points this way because the left side is the probability of being in one of the *lowest*  $k$  states. Smaller lower-tail probabilities mean more mass on high skill.

## 2.4 ARA and a derivative identity

For  $p' > 0$ , define

$$A(x) := -\frac{p''(x)}{p'(x)}.$$

Concavity implies  $A(x) \geq 0$ . IARA means  $A$  increases; DARA means it decreases. Equivalently,  $A(x) = -(\log p'(x))'$ , so ARA is the instantaneous proportional rate at which marginal productivity falls.

## 2.5 Modes from adjacent probability ratios

For any strictly positive discrete distribution,

$$r_k := \frac{\pi_{k+1}}{\pi_k}$$

determines local direction:  $r_k > 1$  means the mass rises from  $k$  to  $k + 1$ , while  $r_k < 1$  means it falls. A decreasing sequence  $r_k$  crosses 1 at most once, hence gives a unimodal distribution. Multiple crossings of 1 allow multiple peaks. This observation is the bridge from effort non-monotonicity to skill polarization.

# 3 The deterministic basic model

## 3.1 Primitives and the agent's problem

The agent has skill  $s \geq 0$ , chooses effort  $e \geq 0$ , and receives deterministic AI assistance  $a \geq 0$ . Total input and utility are

$$x = s + e + a, \quad u(e; s, a) = p(s + e + a) - \gamma e.$$

Productivity excludes effort cost:

$$p^*(s, a) = p(s + e^*(s, a) + a).$$

This distinction matters: utility is the choice criterion; productivity is the outcome the paper evaluates.

## 3.2 Full solution of the optimization problem

Set  $x = s + e + a$ . Because  $e = x - s - a$  and  $e \geq 0$  iff  $x \geq s + a$ ,

$$\max_{e \geq 0} \{p(s + e + a) - \gamma e\} = \max_{x \geq s + a} \{p(x) - \gamma(x - s - a)\}$$

$$= \gamma(s + a) + \max_{x \geq s+a} \{p(x) - \gamma x\}.$$

The first term is constant in  $x$ . Thus we maximize the concave function  $g(x) = p(x) - \gamma x$  over  $[s + a, \infty)$ . There are two cases.

**Case 1:**  $s + a \leq x^*$ . The unconstrained largest maximizer is feasible, so  $x = x^*$  and

$$e^* = x^* - s - a, \quad p^* = p(x^*).$$

**Case 2:**  $s + a > x^*$ . To the right of the largest maximizer, concavity makes  $g$  weakly decreasing. The constrained maximum is the left endpoint  $x = s + a$ , so

$$e^* = 0, \quad p^* = p(s + a).$$

Combining the cases gives the paper's Proposition 2.1:

Proposition 3.1 (Deterministic optimizer).

$$\boxed{e^*(s, a) = (x^* - s - a)_+}, \quad \boxed{p^*(s, a) = \max\{p(x^*), p(s + a)\}}.$$

**Checkpoint.** When  $s + a < x^*$ , an additional unit of AI reduces effort by exactly one unit and leaves total input and productivity unchanged. When  $s + a > x^*$ , effort is already zero and AI raises total input directly.

### 3.3 A concrete example

Take  $p(x) = \alpha \log(1 + x)$ ,  $0 < \gamma < \alpha$ . The FOC  $p'(x) = \alpha/(1 + x) = \gamma$  gives

$$x^* = \frac{\alpha}{\gamma} - 1.$$

Hence

$$e^*(s, a) = \left( \frac{\alpha}{\gamma} - 1 - s - a \right)_+, \quad p^*(s, a) = \alpha \log \left( 1 + \max \left\{ \frac{\alpha}{\gamma} - 1, s + a \right\} \right).$$

This example makes the kink at  $s + a = x^*$  transparent.

## 4 Endogenous skill development and the first paradox

### 4.1 The continuous-time skill process

Let  $S = (s_1, \dots, s_N)$  with  $s_1 \leq \dots \leq s_N$ . At state  $s_k$ , the myopic agent chooses

$$e_k(a) := e^*(s_k, a) = (x^* - s_k - a)_+.$$

The upward and downward transition rates are

$$K_{s_k, s_{k+1}}(a) = \lambda(e_k(a)), \quad K_{s_k, s_{k-1}}(a) = \mu,$$

where  $\lambda > 0$  is increasing, concave, and differentiable, and  $\mu > 0$  is skill decay.

Detailed balance gives

Proposition 4.1 (Stationary skill distribution).

$$\pi_k(a) = \pi_1(a) \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu},$$

$$\pi_1(a) = \left( 1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_i(a))}{\mu} \right)^{-1}.$$

The stationary effort and productivity are

$$\mathcal{E}(a) = \sum_{k=1}^N \pi_k(a) e_k(a), \quad \mathcal{P}(a) = \sum_{k=1}^N \pi_k(a) p^*(s_k, a).$$

## 4.2 Why detailed balance is valid here

Write the stationarity equation  $\pi Q = 0$ . At an interior state  $k$  it is

$$\pi_{k-1} b_{k-1} + \pi_{k+1} \mu - \pi_k (b_k + \mu) = 0.$$

If the edge flows balance,  $\pi_{k-1} b_{k-1} = \pi_k \mu$  and  $\pi_k b_k = \pi_{k+1} \mu$ , then the inflow equals the outflow term by term. The boundary equations also hold. Positivity of all admissible rates makes the finite chain irreducible, so the normalized stationary distribution is unique.

## 4.3 Activity regimes

Define

$$I_m = ((x^* - s_{m+1})_+, (x^* - s_m)_+], \quad m = 1, \dots, N-1,$$

with  $I_0 = [(x^* - s_1)_+, \infty)$  and  $I_N = [0, (x^* - s_N)_+]$ . For  $a$  in the interior of  $I_m$ :

- states  $k \leq m$  exert positive effort, so  $e_k = x^* - s_k - a$  and  $p_k^* = p(x^*)$ ;
- states  $k \geq m+1$  exert zero effort, so  $e_k = 0$  and  $p_k^* = p(s_k + a)$ .

The partition matters because the formula is smooth inside each  $I_m$  but changes at its boundaries.

## 4.4 The sensitivity gap

For  $m \in \{1, \dots, N-1\}$  define

$$\Delta_m = \sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)} - \frac{p'(x^* + s_{m+1} - s_m)}{p(x^* + s_{m+1} - s_m) - p(x^*)}.$$

Each term is a semi-elasticity. The first is the proportional response of the product of upward rates to an effort change. The second is the proportional response of the productivity premium of the first inactive state to additional AI. A positive gap says the distributional deskilling channel is locally more sensitive.



#### 4.5 A complete proof of skill degradation

Let  $a_\ell < a_h$ . The deterministic solution implies  $e_i(a_\ell) \geq e_i(a_h)$  for every state. Since  $\lambda$  is increasing,

$$r_i^\ell := \frac{\lambda(e_i(a_\ell))}{\mu} \geq r_i^h := \frac{\lambda(e_i(a_h))}{\mu}.$$

Define unnormalized weights  $w_1 = 1$  and  $w_k = \prod_{i=1}^{k-1} r_i$ . We must show

$$\frac{\sum_{j=1}^k w_j^\ell}{\sum_{j=1}^N w_j^\ell} \leq \frac{\sum_{j=1}^k w_j^h}{\sum_{j=1}^N w_j^h}.$$

Cross-multiplying, cancel the common low–low block. What remains is the claim that, for  $j \leq k < n$ ,

$$w_n^\ell w_j^h \geq w_j^\ell w_n^h.$$

Indeed,

$$\frac{w_n^\ell}{w_j^\ell} = \prod_{i=j}^{n-1} r_i^\ell \geq \prod_{i=j}^{n-1} r_i^h = \frac{w_n^h}{w_j^h}.$$

Summing these pairwise inequalities proves the lower-tail inequality.

**Proposition 4.2 (Skill degradation).** *The low-assistance stationary distribution first-order stochastically dominates the high-assistance one:*

$$\sum_{i=1}^k \pi_i(a_\ell) \leq \sum_{i=1}^k \pi_i(a_h), \quad k = 1, \dots, N.$$

**Intuition.** At every current skill, more AI reduces effort. This lowers every upward transition rate without changing the decay rate. Stationary mass therefore shifts toward low skill. This is the distributional ingredient of the Simpson-type reversal.

#### 4.6 Rewriting stationary productivity on one regime

Fix  $a \in I_m$ ,  $1 \leq m \leq N - 1$ , and define

$$A_{k,m}(a) := \left[ \prod_{i=k}^m \lambda(x^* - s_i - a) \right]^{-1}, \quad B_{k,m} := \lambda(0)^{k-m-1}.$$

Start from the stationary weights, split active and inactive states, and divide numerator and denominator by  $\prod_{i=1}^m \lambda(x^* - s_i - a)$ . This gives

$$\mathcal{P}(a) = p(x^*) + \frac{N_m(a)}{D_m(a)},$$

where

$$N_m(a) = \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} [p(s_k + a) - p(x^*)],$$

$$D_m(a) = \sum_{k=1}^m \frac{A_{k,m}(a)}{\mu^{k-1}} + \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}}.$$

Differentiate by the quotient rule:

$$\mathcal{P}'(a) = \frac{1}{D_m(a)^2} \left[ \left( \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} p'(s_k + a) \right) D_m(a) - N_m(a) \left( \sum_{k=1}^m \frac{A'_{k,m}(a)}{\mu^{k-1}} \right) \right]. \quad (1)$$

The first term is the direct productivity gain. The second is the loss caused by the endogenous shift in stationary weights.

#### 4.7 Why the derivative crosses zero at most once

Let

$$h_m(a) := \frac{\sum_{k=1}^m A'_{k,m}(a)/\mu^{k-1}}{\sum_{k=1}^m A_{k,m}(a)/\mu^{k-1} + \sum_{k=m+1}^N B_{k,m}/\mu^{k-1}}.$$

After dividing the numerator of (1) by  $D_m(a)$ , its sign is the sign of

$$G_m(a) = \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} p'(s_k + a) - \sum_{k=m+1}^N \frac{B_{k,m}}{\mu^{k-1}} [p(s_k + a) - p(x^*)] h_m(a).$$

The first sum is decreasing by concavity of  $p$ . The bracketed productivity-premium sum is increasing. It remains to show  $h_m$  is increasing.

Put  $\bar{A}_k = A_{k,m}/\mu^{k-1}$  and  $\bar{B} = \sum_{k=m+1}^N B_{k,m}/\mu^{k-1}$ . If  $L_k = \sum_{j=k}^m \lambda'(x^* - s_j - a)/\lambda(x^* - s_j - a)$ , then

$$\bar{A}'_k = \bar{A}_k L_k,$$

and concavity of  $\lambda$  gives

$$\bar{A}''_k = \bar{A}_k L_k^2 + \bar{A}_k \sum_{j=k}^m \frac{(\lambda')^2 - \lambda''\lambda}{\lambda^2} \geq \bar{A}_k L_k^2.$$

Differentiating  $h_m = (\sum \bar{A}'_k)/(\sum \bar{A}_k + \bar{B})$ , its numerator is

$$(\sum \bar{A}''_k)(\sum \bar{A}_k + \bar{B}) - (\sum \bar{A}'_k)^2.$$

It is positive because

$$(\sum \bar{A}''_k)(\sum \bar{A}_k + \bar{B}) > (\sum \bar{A}_k L_k^2)(\sum \bar{A}_k) \geq (\sum \bar{A}_k L_k)^2$$

by Cauchy–Schwarz. Thus  $h_m$  increases,  $G_m$  decreases, and  $\mathcal{P}'$  crosses zero at most once.

**Lemma 4.3** (Within-regime shape). *On  $I_m$ , stationary productivity is either increasing everywhere or increasing and then decreasing. A decreasing subregion exists exactly when the left derivative at the right endpoint is negative.*

#### 4.8 The large-decay asymptotic and Theorem 1

Evaluate the numerator of (1) at the right endpoint  $a = x^* - s_m$ . Express it as a polynomial in  $z = 1/\mu$ . Its lowest nonzero order is  $z^m$ . Only the pair consisting of the lowest active weight and the first inactive state survives at that order. Its coefficient is

$$\frac{1}{\prod_{i=1}^m \lambda(s_m - s_i)} \left[ p'(x^* + s_{m+1} - s_m) \right]$$

$$\begin{aligned}
& - (p(x^* + s_{m+1} - s_m) - p(x^*)) \sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)} \Big] \\
& = -C_m \Delta_m,
\end{aligned}$$

where

$$C_m = \frac{p(x^* + s_{m+1} - s_m) - p(x^*)}{\prod_{i=1}^m \lambda(s_m - s_i)} > 0.$$

As  $\mu \rightarrow \infty$ ,  $z \rightarrow 0$ , so the sign of the whole polynomial equals the sign of its lowest nonzero coefficient. Therefore the endpoint derivative is negative if  $\Delta_m > 0$  and positive if  $\Delta_m < 0$ , once  $\mu$  is large enough.

Theorem 4.4 (Paper’s Theorem 1). *There is  $\bar{\mu} > 0$  such that, for every  $\mu > \bar{\mu}$  and nonempty  $I_m$ :*

1. *if  $1 \leq m \leq N - 1$  and  $\Delta_m > 0$ ,  $\mathcal{P}$  increases and then decreases on  $I_m$ ;*
2. *if  $m \in \{0, N\}$  or  $\Delta_m < 0$ ,  $\mathcal{P}$  is weakly increasing on  $I_m$ .*

For  $I_N$ , every state exerts enough effort to keep total input at  $x^*$ , so  $\mathcal{P} = p(x^*)$ . For  $I_0$ , no state exerts effort and every state-level productivity is increasing. These boundary regimes do not need the sensitivity-gap argument.

#### 4.9 Exact two-state calculation

Let  $N = 2$  and  $s_1 < s_2$ . On  $I_1$ ,

$$L(a) := \lambda(x^* - s_1 - a), \quad \mathcal{P}(a) = \frac{\mu p(x^*) + L(a)p(s_2 + a)}{\mu + L(a)}.$$

Differentiation gives

$$\mathcal{P}'(a) = \frac{(\mu + L)Lp'(s_2 + a) - \mu\lambda'(x^* - s_1 - a)[p(s_2 + a) - p(x^*)]}{(\mu + L)^2}.$$

The numerator decreases in  $a$ . At the left endpoint it is positive. At the right endpoint  $a = x^* - s_1$  it is negative exactly when

$$\mu > \bar{\mu}_2 := \frac{\lambda(0)^2 p'(x^* + s_2 - s_1)}{\lambda'(0)[p(x^* + s_2 - s_1) - p(x^*)] - \lambda(0)p'(x^* + s_2 - s_1)},$$

and the denominator defining  $\bar{\mu}_2$  is positive. Dividing that positivity condition by  $\lambda(0)[p(x^* + s_2 - s_1) - p(x^*)]$  shows it is exactly  $\Delta_1 > 0$ .

Thus the two-state model has an  $N$ -shaped global profile: flat/increasing at very low  $a$ , then increasing–decreasing on  $I_1$ , then increasing again after all effort is crowded out.

#### 4.10 Why an arbitrarily large shortfall is possible

The paper constructs a nearly linear concave production function with a kink at  $x^*$ , makes  $\lambda(0)$  close to zero, and compares  $a_h = x^*$  to  $a_\ell = x^* - z$ . At  $a_h$ , no state exerts effort and the chain remains near the lowest state. At  $a_\ell$ , only the lowest state exerts effort  $z$ ; choosing  $\lambda(z)/\mu$  large moves mass toward  $s_2$ .

In the limiting calculation,

$$\frac{\mathcal{P}(x^*)}{\mathcal{P}(x^* - z)} \longrightarrow \frac{p(x^*)}{\frac{1}{1 + \lambda(z)/\mu} \left[ p(x^*) + \frac{\lambda(z)}{\mu} p(x^* + s_2 - z) \right]}.$$

Taking the production function arbitrarily close to linear and then setting  $x^* = z = \varepsilon s_2/4$ , with  $\lambda(z)/(\mu + \lambda(z)) \geq 1 - \varepsilon/4$ , makes this ratio at most  $\varepsilon$ . The ordering of limits is important: first choose the almost-linear  $p$ , then small  $\lambda(0)$ , then a steep enough rise to  $\lambda(z)$ .

#### 4.11 Simpson’s paradox and the exogenous benchmark

At fixed  $s_k$ ,  $p^*(s_k, a)$  never falls. Nevertheless,

$$\mathcal{P}(a) = \sum_k \pi_k(a) p^*(s_k, a)$$

can fall because the weights  $\pi_k(a)$  move toward low-skill states. If transition rates were exogenous,  $\pi$  would not depend on  $a$  and

$$\mathcal{P}'(a) = \sum_k \pi_k \frac{\partial p^*(s_k, a)}{\partial a} \geq 0.$$

Thus endogeneity of the weights, not heterogeneity alone, is essential.

### 5 Unreliable AI and effort cannibalization

#### 5.1 Timing and expected utility

AI assistance is binary:

$$a = \begin{cases} \bar{a}, & \text{with probability } q, \\ 0, & \text{with probability } 1 - q. \end{cases}$$

The agent chooses effort before observing the realization. Expected utility and realized expected productivity at the optimizer are

$$\begin{aligned} u(e; s, \bar{a}, q) &= q, p(s + e + \bar{a}) + (1 - q)p(s + e) - \gamma e, \\ e^*(s, \bar{a}, q) &\in \arg \max_{e \geq 0} u(e; s, \bar{a}, q), \\ p^*(s, \bar{a}, q) &= q, p(s + e^* + \bar{a}) + (1 - q)p(s + e^*). \end{aligned}$$

The commitment assumption is the source of the second paradox. If effort could be adapted perfectly after observing  $a$ , the deterministic solution would apply state by state.

#### 5.2 Interior versus corner effort

The marginal utility of effort is

$$u_e = q, p'(s + e + \bar{a}) + (1 - q)p'(s + e) - \gamma.$$

It decreases in  $e$  by concavity. Therefore:

- if  $qp'(s + \bar{a}) + (1 - q)p'(s) > \gamma$ , the optimum is interior and solves

$$q, p'(s + e^* + \bar{a}) + (1 - q)p'(s + e^*) = \gamma; \tag{2}$$

- if  $qp'(s + \bar{a}) + (1 - q)p'(s) < \gamma$ , then  $u_e(0) < 0$  and  $e^* = 0$ .

Equality is a boundary/tie case and can be handled by one-sided limits.

### 5.3 Implicit differentiation, without skipped algebra

Assume the interior case and abbreviate

$$x_0 = s + e^*, \quad x_1 = s + e^* + \bar{a}, \quad D = qp''(x_1) + (1 - q)p''(x_0) < 0.$$

Differentiate (2) with respect to  $\bar{a}$ :

$$qp''(x_1) \left(1 + \frac{\partial e^*}{\partial \bar{a}}\right) + (1 - q)p''(x_0) \frac{\partial e^*}{\partial \bar{a}} = 0.$$

Collect the two terms containing the derivative:

$$D \frac{\partial e^*}{\partial \bar{a}} + qp''(x_1) = 0,$$

so

$$\boxed{\frac{\partial e^*}{\partial \bar{a}} = -\frac{qp''(x_1)}{D}}. \quad (3)$$

Both numerator and denominator are nonpositive, so the fraction inside the minus sign is nonnegative; therefore  $\partial e^*/\partial \bar{a} \leq 0$ . More proficient AI crowds out effort even though it is unreliable.

Now differentiate productivity:

$$\begin{aligned} \frac{\partial p^*}{\partial \bar{a}} &= qp'(x_1) (1 + e_a^*) + (1 - q)p'(x_0)e_a^* \\ &= qp'(x_1) + [qp'(x_1) + (1 - q)p'(x_0)]e_a^*. \end{aligned}$$

Substitute (3):

$$\begin{aligned} \frac{\partial p^*}{\partial \bar{a}} &= qp'(x_1) - [qp'(x_1) + (1 - q)p'(x_0)] \frac{qp''(x_1)}{D} \\ &= \frac{qp'(x_1)D - qp''(x_1)[qp'(x_1) + (1 - q)p'(x_0)]}{D} \\ &= \frac{q(1 - q)[p'(x_1)p''(x_0) - p''(x_1)p'(x_0)]}{D}. \end{aligned}$$

Factor  $p'(x_0)p'(x_1)$  and use  $A(x) = -p''(x)/p'(x)$ :

$$\boxed{\frac{\partial p^*}{\partial \bar{a}} = \frac{q(1 - q)p'(x_0)p'(x_1)}{D} [A(x_1) - A(x_0)]}. \quad (4)$$

The prefactor outside the square bracket is negative because  $D < 0$ . Hence

$$\text{sgn} \left( \frac{\partial p^*}{\partial \bar{a}} \right) = \text{sgn}(-[A(x_1) - A(x_0)]).$$

At the corner  $e^* = 0$ ,

$$p^* = qp(s + \bar{a}) + (1 - q)p(s), \quad \frac{\partial p^*}{\partial \bar{a}} = qp'(s + \bar{a}) \geq 0.$$

**Intuition.** Equation (4) contains both forces. A larger  $\bar{a}$  directly helps in the good state, but it reduces effort in both states. IARA makes the good-state marginal product deteriorate proportionally faster than the bad-state marginal product, so the effort reduction can dominate while effort remains interior.

## 5.4 The IARA–DARA theorem

Assume  $s < x^*$  and  $0 < q < 1$ . Since  $p'$  decreases, the function

$$H(\bar{a}) = qp'(s + \bar{a}) + (1 - q)p'(s)$$

decreases in  $\bar{a}$ . Define the possibly infinite threshold

$$\tau := \sup\{z \geq 0 : H(z) \geq \gamma\}.$$

For  $\bar{a} < \tau$ , effort is positive and (4) applies. For  $\bar{a} > \tau$ , effort is zero and the derivative is positive.

**Theorem 5.1** (Paper’s Theorem 2). *For concave  $p$ , linear effort cost,  $s < x^*$ , and  $0 < q < 1$ :*

1. *if  $p$  satisfies IARA,  $p^*(s, \bar{a}, q)$  first decreases and then possibly increases in  $\bar{a}$ ;*
2. *if  $p$  satisfies DARA,  $p^*(s, \bar{a}, q)$  increases in  $\bar{a}$ .*

For IARA,  $x_1 > x_0$  implies  $A(x_1) > A(x_0)$ , so the interior derivative is negative; after the corner is reached, it is positive. For DARA,  $A(x_1) < A(x_0)$ , so even the interior derivative is positive. If  $(1 - q)p'(s) \geq \gamma$ , then  $H(\bar{a}) \geq \gamma$  for all  $\bar{a}$ , so  $\tau = \infty$  and the increasing region never arrives.

## 5.5 Examples of ARA classes

| $p(x)$                     | $A(x)$                  | Class                      |
|----------------------------|-------------------------|----------------------------|
| $\log(1 + cx)$             | $c/(1 + cx)$            | DARA                       |
| $x^\alpha, 0 < \alpha < 1$ | $(1 - \alpha)/x$        | DARA                       |
| $1 - e^{-bx}$              | $b$                     | CARA (weak IARA and DARA)  |
| $1 - e^{-bx^c}, c > 1$     | $bcx^{c-1} - (c - 1)/x$ | IARA on its concave domain |
| $\int_0^x e^{-t^2} dt$     | $2x$                    | IARA                       |

For the last line,  $p'(x) = e^{-x^2}$  and  $p''(x) = -2xe^{-x^2}$ , so  $A(x) = 2x$  immediately.

## 5.6 Piecewise-linear production: exact optimizer

Let

$$p(x) = \min\{1, \beta x\}, \quad \beta > \gamma.$$

There are three regions, depending on whether neither, one, or both realizations have reached the cap:

$$u(e) = \begin{cases} (\beta - \gamma)e + \beta s + q\beta\bar{a}, & s + e + \bar{a} \leq 1/\beta, \\ [(1 - q)\beta - \gamma]e + (1 - q)\beta s + q, & s + e \leq 1/\beta < s + e + \bar{a}, \\ 1 - \gamma e, & s + e > 1/\beta. \end{cases}$$

The first slope is positive. The third is negative. The middle slope is nonnegative exactly when

$$(1 - q)\beta - \gamma \geq 0 \iff q \leq \frac{\beta - \gamma}{\beta}.$$

Therefore

$$e^*(s, \bar{a}, q) = \begin{cases} (1/\beta - s)_+, & q \leq (\beta - \gamma)/\beta, \\ (1/\beta - s - \bar{a})_+, & q > (\beta - \gamma)/\beta, \end{cases}$$

$$p^*(s, \bar{a}, q) = \begin{cases} 1, & q \leq (\beta - \gamma)/\beta, \\ q + (1 - q) \max\{\min(1, \beta s), 1 - \beta \bar{a}\}, & q > (\beta - \gamma)/\beta. \end{cases}$$

### 5.7 Arbitrarily small productivity ratio

Set  $s = 0$ ,  $a_\ell = 0$ , and  $a_h = 1/\beta$ . Choose  $q > (\beta - \gamma)/\beta$ . The high-reliability branch above gives

$$p^*(0, 0, q) = 1, \quad p^*(0, 1/\beta, q) = q.$$

Given any  $\varepsilon > 0$ , take  $q = \varepsilon$  and choose  $\gamma$  so that  $(\beta - \gamma)/\beta = \varepsilon/2$ . Then the required strict inequality holds and

$$\frac{p^*(0, a_h, q)}{p^*(0, a_\ell, q)} = \varepsilon.$$

The construction is capped and kinked; the paper also explains how a small smooth perturbation preserves an arbitrarily small ratio under relaxed IARA.

### 5.8 Certainty equivalent and the hazard-rate connection

Define  $a_{CE}$  implicitly by

$$p(s + e + a_{CE}) = qp(s + e + \bar{a}) + (1 - q)p(s + e).$$

A second-order Arrow–Pratt approximation gives

$$q\bar{a} - a_{CE} \approx \frac{1}{2}A(s + e + q\bar{a})q(1 - q)\bar{a}^2.$$

Under IARA, more effort raises the risk premium and lowers the effective contribution of AI. This is an interpretation of realized expected production, not a separate optimization problem.

If  $p(x) = \mathbb{E}[\min\{X, x\}]$  for a nonnegative  $X$  with density  $f$  and CDF  $F$ , then

$$p'(x) = 1 - F(x), \quad p''(x) = -f(x), \quad A(x) = \frac{f(x)}{1 - F(x)}.$$

Thus IARA is exactly increasing hazard rate for this production representation.

### 5.9 Perfect adaptation benchmark

If the agent observes the realization before choosing effort, then

$$e^*(s, a) = (x^* - s - a)_+.$$

Expected productivity becomes

$$q \max\{p(x^*), p(s + \bar{a})\} + (1 - q) \max\{p(x^*), p(s)\},$$

which is weakly increasing in  $\bar{a}$ . Therefore imperfect adaptation, not randomness by itself, is essential for Theorem 2.

## 6 AI literacy and skill polarization

### 6.1 Signal technology and Bayes' rule

The AI realization remains  $a \in \{0, \bar{a}\}$  with prior  $\mathbb{P}(a = \bar{a}) = q$ . The agent observes  $\hat{a} \in \{0, \bar{a}\}$  with skill-dependent accuracy

$$\mathbb{P}(\hat{a} = a \mid s) = v(s), \quad v(s) \in [1/2, 1],$$

where  $v$  is increasing and concave.

For a positive signal, Bayes' rule gives

$$\begin{aligned} q_1(q, s) &= \mathbb{P}(a = \bar{a} \mid \hat{a} = \bar{a}, s) \\ &= \frac{\mathbb{P}(\hat{a} = \bar{a} \mid a = \bar{a}, s) \mathbb{P}(a = \bar{a})}{\mathbb{P}(\hat{a} = \bar{a} \mid s)} \\ &= \frac{qv(s)}{qv(s) + (1-q)[1-v(s)]}. \end{aligned}$$

For a zero signal,

$$q_0(q, s) = \frac{q[1-v(s)]}{q[1-v(s)] + (1-q)v(s)}.$$

Signal probabilities are

$$\begin{aligned} \rho_1(s) &:= \mathbb{P}(\hat{a} = \bar{a} \mid s) = qv(s) + (1-q)[1-v(s)] = (2q-1)v(s) + 1-q, \\ \rho_0(s) &:= \mathbb{P}(\hat{a} = 0 \mid s) = q[1-v(s)] + (1-q)v(s) = (1-2q)v(s) + q. \end{aligned}$$

After observing the signal, a Bayesian signal follower uses the ex-ante optimizer from Section 4 with the corresponding posterior. Expected effort is

$$e_b^*(s, \bar{a}, q) = \rho_1(s)e^*(s, \bar{a}, q_1(q, s)) + \rho_0(s)e^*(s, \bar{a}, q_0(q, s)).$$

### 6.2 Posterior monotonicity

Let  $v = v(s)$ . Differentiating with respect to  $v$  gives

$$\begin{aligned} \frac{\partial q_1}{\partial v} &= \frac{q(1-q)}{[qv + (1-q)(1-v)]^2} > 0, \\ \frac{\partial q_0}{\partial v} &= -\frac{q(1-q)}{[q(1-v) + (1-q)v]^2} < 0. \end{aligned}$$

Since  $v'(s) \geq 0$ ,  $q_1$  rises and  $q_0$  falls with skill. If  $v(0) = 1/2$ , then both posteriors begin at  $q$ :  $q_1(q, 0) = q_0(q, 0) = q$ .

### 6.3 The capped-linear threshold

Continue with  $p(x) = \min\{1, \beta x\}$  and  $\beta > \gamma$ . Define the reliability cutoff

$$r^* := \frac{\beta - \gamma}{\beta}$$

and the marginal return gap

$$\omega = (\beta - \gamma) - q\beta = \beta(r^* - q).$$



Thus  $\omega < 0$  means  $q > r^*$  and  $\omega \geq 0$  means  $q \leq r^*$ .

Solving  $q_0(q, s) = r^*$  gives, after cross-multiplication,

$$\begin{aligned} q(1-v) &= r^*[q(1-v) + (1-q)v], \\ q(1-r^*)(1-v) &= r^*(1-q)v, \\ \frac{v}{1-v} &= \frac{q}{1-q} \frac{1-r^*}{r^*} = \frac{q}{1-q} \frac{\gamma}{\beta-\gamma}. \end{aligned}$$

Hence when  $\omega < 0$  the crossing skill is

$$\tilde{s} = v^{-1} \left( \frac{\frac{q}{1-q} \frac{\gamma}{\beta-\gamma}}{1 + \frac{q}{1-q} \frac{\gamma}{\beta-\gamma}} \right).$$

Similarly, solving  $q_1 = r^*$  gives, for  $\omega \geq 0$ ,

$$\tilde{s} = v^{-1} \left( \frac{1}{1 + \frac{q}{1-q} \frac{\gamma}{\beta-\gamma}} \right).$$

#### 6.4 Piecewise expected effort: the AI-critical case

Suppose  $\omega < 0$ . Before  $q_0$  crosses the cutoff, both posteriors exceed  $r^*$  and both conditional efforts use the low-effort branch. After the crossing, the zero-signal posterior uses the high-effort branch:

$$e_b^*(s, \bar{a}, q) = \begin{cases} (1/\beta - s - \bar{a})_+, & s < \tilde{s}, \\ \rho_1(s) (1/\beta - s - \bar{a})_+ + \rho_0(s) (1/\beta - s)_+, & s \geq \tilde{s}. \end{cases}$$

If  $\tilde{s} < 1/\beta$  and  $\bar{a} > 0$ , the function jumps upward at  $\tilde{s}$  because the zero-signal conditional effort changes from  $(1/\beta - \tilde{s} - \bar{a})_+$  to  $(1/\beta - \tilde{s})_+$ . Effort decreased just before the threshold but rises at it: non-monotonicity. If  $\tilde{s} \geq 1/\beta$ , both terms are already zero at the threshold, so no upward jump is possible.

#### 6.5 Piecewise expected effort: the effort-critical case

Suppose  $\omega \geq 0$ . Before  $q_1$  crosses the cutoff, both posterior choices use the high-effort branch. After the crossing,

$$e_b^*(s, \bar{a}, q) = \begin{cases} (1/\beta - s)_+, & s < \tilde{s}, \\ \rho_1(s) (1/\beta - s - \bar{a})_+ + \rho_0(s) (1/\beta - s)_+, & s \geq \tilde{s}. \end{cases}$$

If  $q \geq 1/2$ , then  $\rho_1$  rises with skill and shifts weight toward the smaller effort term, while both terms themselves fall. Expected effort must fall. Non-monotonicity is possible only if  $q < 1/2$ , because then  $\rho_0(s) = (1-2q)v(s) + q$  increases and shifts weight toward the larger effort term.

Assume  $q < 1/2$ . Where both positive-part terms are active,

$$e_b^* = \frac{1}{\beta} - s + \bar{a}(1-2q)v(s) - (1-q)\bar{a},$$

so

$$\frac{\partial e_b^*}{\partial s} = -1 + \bar{a}(1-2q)v'(s).$$

Where only the zero-signal term is active,

$$e_b^* = \rho_0(s)(1/\beta - s),$$

so

$$\frac{\partial e_b^*}{\partial s} = (1 - 2q)v'(s)(1/\beta - s) - [(1 - 2q)v(s) + q].$$

Concavity of  $v$  makes each derivative decreasing within its region, so it suffices to inspect boundary values. A positive derivative can occur exactly when

$$\frac{1}{\beta} - \tilde{s} > \frac{(1 - 2q)v(\tilde{s}) + q}{(1 - 2q)v'(\tilde{s})}.$$

## 6.6 The paper’s polarization condition

Combining the two cases yields Condition 5.3:

Condition 6.1 (Polarization condition).

$$\tilde{s} < \frac{1}{\beta} - \frac{(1 - 2q)v(\tilde{s}) + q}{(1 - 2q)v'(\tilde{s})} \mathbb{I}\{\omega \geq 0\},$$

and either (a)  $\omega < 0$ , or (b)  $\omega \geq 0$  and  $q < 1/2$ .

For  $\omega < 0$ , the indicator removes the derivative term and the condition is simply  $\tilde{s} < 1/\beta$ . For  $\omega \geq 0$ , it requires enough remaining distance to the production cap for the skill-induced shift in signal probabilities to overcome the direct decline in effort.

## 6.7 Feeding expected effort into skill dynamics

Upward and downward rates are now

$$K_{s_k, s_{k+1}} = \lambda(e_b^*(s_k, \bar{a}, q)), \quad K_{s_k, s_{k-1}} = \mu.$$

The stationary distribution is

$$\begin{aligned} \pi_k(\mu, \bar{a}, q, S) &= \pi_1 \prod_{i=1}^{k-1} \frac{\lambda(e_b^*(s_i, \bar{a}, q))}{\mu}, \\ \pi_1 &= \left[ 1 + \sum_{k=2}^N \prod_{i=1}^{k-1} \frac{\lambda(e_b^*(s_i, \bar{a}, q))}{\mu} \right]^{-1}. \end{aligned}$$

The adjacent ratio is the essential simplification:

$$\boxed{\frac{\pi_{k+1}}{\pi_k} = \frac{\lambda(e_b^*(s_k, \bar{a}, q))}{\mu}}. \tag{5}$$

## 6.8 From effort shape to distribution shape

**Lemma 6.2** (Decreasing effort implies unimodality). *If  $e_b^*(s, \bar{a}, q)$  decreases in  $s$  and  $\lambda$  increases, the ratios in (5) decrease in  $k$ . They cross 1 at most once. Thus  $\pi_k$  first rises and then falls, possibly with a flat top: the distribution is unimodal.*

**Lemma 6.3** (Non-monotone effort can generate multimodality). *If  $e_b^*(s, \bar{a}, q)$  is non-monotone, then an interval exists on which  $\lambda(e_b^*(s, \bar{a}, q))$  has a local valley or hill. Choose  $\mu$  strictly between an extremal value and nearby values so that  $\lambda(e_b^*)/\mu$  crosses 1 more than once. Select a finite, sufficiently rich ordered skill grid containing points from each sign region. By (5),  $\pi$  then falls–rises–falls (or the corresponding alternating pattern), creating at least two local maxima.*

The second lemma is existential: the theorem does not assert multimodality for every grid or every decay rate. It asserts that some  $\mu$  and sufficiently rich  $S$  reveal the non-monotonic effort shape.

## 6.9 The polarization theorem

Theorem 6.4 (Paper’s Theorem 3). *Let  $p(x) = \min\{1, \beta x\}$ ,  $c(e) = \gamma e$ ,  $\beta > \gamma$ , and let  $\lambda$  be increasing. Suppose  $v : [0, \infty) \rightarrow [1/2, 1]$  is increasing and concave and  $0 < q < 1$ .*

1. *If the polarization condition holds, there is  $\tilde{a} \geq 0$  such that for  $\bar{a} > \tilde{a}$  some  $\mu > 0$  and skill grid  $S$  yield a multimodal stationary distribution; for  $\bar{a} \leq \tilde{a}$ , every  $\mu$  and  $S$  yield a unimodal distribution.*
2. *If the condition fails, the distribution is unimodal for every  $\bar{a}$ ,  $\mu$ , and  $S$ .*

The proof is now only composition: Condition 5.3 is equivalent to the effort non-monotonicity characterized above; the two adjacent-ratio lemmas translate effort shape into distribution shape. In the  $\omega < 0$  case,  $\tilde{a} = 0$  because any positive proficiency creates the threshold jump.

## 6.10 Constant verification as a falsification benchmark

If  $v(s) \equiv v$ , then  $q_1, q_0, \rho_1, \rho_0$  do not depend on skill. For any concave  $p$ , the ex-ante optimizer  $e^*(s, \bar{a}, r)$  decreases in  $s$ : increasing  $s$  shifts down the marginal-return schedule

$$rp'(s + e + \bar{a}) + (1 - r)p'(s + e).$$

Expected effort is a fixed convex combination of two decreasing functions, hence decreases. By the adjacent-ratio lemma, the stationary distribution must be unimodal. Skill-dependent verification is therefore essential; simply giving everyone the same imperfect signal cannot polarize skill in this model.

## 7 Synthesis: what each paradox requires

| Phenomenon                | Direct effect of AI                                | Endogenous counterforce                                                                              | Condition selecting the sign                              |
|---------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Skill-development paradox | Higher $a$ weakly raises state-level productivity. | Lower effort reduces upward skill transitions and shifts stationary mass downward.                   | Sensitivity gap $\Delta_m$ and sufficiently large $\mu$ . |
| Unreliability paradox     | A larger $\bar{a}$ helps in the good realization.  | Ex-ante effort falls in both realizations.                                                           | IARA versus DARA, through $A(x_1) - A(x_0)$ .             |
| Skill polarization        | Better signals can permit adaptive effort.         | Signal accuracy rises with skill, so effort can rise with skill and reinforce the initial advantage. | Condition 5.3 and a sufficiently rich skill grid.         |

The common structure is a derivative or ratio decomposition:

$$\text{net AI effect} = \text{direct gain} - \text{endogenous effort loss}.$$

Section 3 embeds the loss in stationary weights; Section 4 embeds it in an implicit optimizer; Section 5 lets heterogeneity in signal quality reverse the usual effort–skill ordering.

## 8 A graduate teaching sequence

### 8.1 Suggested four-lecture plan

1. **Lecture 1: primitives and deterministic benchmark.** Concavity,  $x^*$ , positive-part optimizer, and the distinction between utility and productivity. End with detailed balance.
2. **Lecture 2: endogenous skill.** Derive the stationary distribution, prove FOSD, construct  $I_m$ , derive (1), and finish with the two-state threshold before the general asymptotic.
3. **Lecture 3: unreliability.** Put the FOC and (4) on the board in full. Use capped-linear production as a sharp example and contrast commitment with perfect adaptation.
4. **Lecture 4: literacy and polarization.** Derive both posteriors, solve the crossing equations, obtain the piecewise effort derivatives, and finish with the adjacent-ratio proof of modes.

### 8.2 Common conceptual mistakes

1. Confusing productivity  $p(\cdot)$  with utility  $p(\cdot) - \gamma e$ .
2. Reading first-order stochastic dominance backwards on an ascending skill grid.
3. Treating  $\mu \rightarrow \infty$  as a high-skill limit; it is a rapid-decay, low-skill limit.
4. Applying the envelope theorem to productivity. The FOC cancels effort effects in utility, not in productivity, which excludes effort cost.
5. Forgetting the corner  $e^* = 0$  in Theorem 2; the derivative formula changes there.
6. Interpreting Theorem 3 as universal multimodality. It is existential in  $(\mu, S)$  above the proficiency threshold.
7. Calling any U- or N-shaped relationship a time-series J-curve. Here the horizontal axis is AI proficiency, not elapsed time after adoption.

## 9 Exercises

- E1.** Let  $p(x) = 2\sqrt{x+1} - 2$  and  $\gamma = 1/2$ . Find  $x^*$ ,  $e^*(s, a)$ , and  $p^*(s, a)$ .
- E2.** In a two-state skill chain, set  $s_1 = 0$ ,  $s_2 = 1$ ,  $x^* = 2$ ,  $\lambda(e) = 1 + e$ , and  $\mu = 2$ . Compute  $\pi(a)$  for  $0 \leq a \leq 1$ .
- E3.** Prove directly that if every upward-to-downward ratio falls, the stationary birth–death distribution shifts downward in the FOSD order.
- E4.** Starting from the unreliable-AI FOC, derive  $e_a^*$  and  $\partial p^*/\partial \bar{a}$  without using the formulas in the text.
- E5.** Classify  $p(x) = \log(1+x)$  and  $p(x) = \int_0^x e^{-t^2} dt$  as IARA or DARA.
- E6.** For capped-linear production, verify the threshold  $r^* = (\beta - \gamma)/\beta$  by comparing the slope of utility in the middle region.
- E7.** Show that  $q_1 \geq q \geq q_0$  when  $v(s) \geq 1/2$ .
- E8.** Suppose the adjacent stationary ratios are  $(0.7, 1.4, 1.3, 0.6)$ . Describe the direction of  $\pi_1, \dots, \pi_5$  and identify its local modes.

**E9.** Explain precisely which implication in the skill-polarization proof fails when  $v(s)$  is constant.

## 10 Solutions to exercises

**S1.**  $p'(x) = 1/\sqrt{x+1}$ . Solving  $p'(x) = 1/2$  gives  $x^* = 3$ . Therefore

$$e^*(s, a) = (3 - s - a)_+, \quad p^*(s, a) = \max\{2, 2\sqrt{s + a + 1} - 2\}.$$

**S2.** For  $0 \leq a \leq 1$ , the low state exerts  $e_1 = 2 - a$  and the high state exerts  $e_2 = 1 - a$ . Only  $e_1$  determines the upward rate from state 1, so

$$\frac{\pi_2}{\pi_1} = \frac{\lambda(e_1)}{\mu} = \frac{3 - a}{2}.$$

Normalization gives

$$\pi_1(a) = \frac{2}{5 - a}, \quad \pi_2(a) = \frac{3 - a}{5 - a}.$$

The derivative of  $\pi_2$  is  $-2/(5 - a)^2 < 0$ , explicitly displaying deskilling.

**S3.** Write  $w_k^x = \prod_{i < k} x_i$  and  $w_k^y = \prod_{i < k} y_i$  with  $x_i \geq y_i$ . After cross-multiplying the two lower-tail fractions, terms with both indices at most  $k$  cancel. For every  $j \leq k < n$ ,  $w_n^x/w_j^x = \prod_{i=j}^{n-1} x_i \geq \prod_{i=j}^{n-1} y_i = w_n^y/w_j^y$ . Summing proves the result.

**S4.** Differentiating the FOC yields

$$e_a^* = -\frac{qp''(x_1)}{qp''(x_1) + (1 - q)p''(x_0)}.$$

Substitute this in  $p_a^* = qp'(x_1)(1 + e_a^*) + (1 - q)p'(x_0)e_a^*$ , combine over the common denominator, and factor the numerator to recover (4).

**S5.** For  $\log(1 + x)$ ,  $A(x) = 1/(1 + x)$ , which decreases: DARA. For the Gaussian integral,  $A(x) = 2x$ , which increases: IARA.

**S6.** In the middle region the good realization is capped and the bad realization is not. The slope is  $(1 - q)\beta - \gamma$ . It is nonnegative iff  $q \leq 1 - \gamma/\beta = (\beta - \gamma)/\beta$ .

**S7.** Posterior odds are

$$\frac{q_1}{1 - q_1} = \frac{q}{1 - q} \frac{v}{1 - v}, \quad \frac{q_0}{1 - q_0} = \frac{q}{1 - q} \frac{1 - v}{v}.$$

Since  $v/(1 - v) \geq 1$ , the first odds exceed the prior odds and the second are below them.

**S8.** The distribution falls from  $\pi_1$  to  $\pi_2$ , rises through  $\pi_3$  and  $\pi_4$ , then falls to  $\pi_5$ . The local modes are  $\pi_1$  at the left boundary and  $\pi_4$  in the interior: a bimodal pattern.

**S9.** With constant  $v$ , both posteriors and both signal weights are constant in skill. Each conditional effort is decreasing in skill, so their fixed convex combination is decreasing. The adjacent ratios then decrease and cannot cross 1 more than once. The failure occurs at the first step: skill cannot change the mixture weights or posterior reliabilities enough to make expected effort non-monotone.

## A Supplementary derivations from Appendices A–C

### A.1 Why a positive sensitivity gap eventually appears

Fix  $\lambda$ ,  $\gamma$ , and  $s_1, \dots, s_m$ . The first term of  $\Delta_m$  is a fixed positive constant

$$L_m = \sum_{j=1}^m \frac{\lambda'(s_m - s_j)}{\lambda(s_m - s_j)}.$$

Let  $d = s_{m+1} - s_m$ . The second term is

$$R(d) = \frac{p'(x^* + d)}{p(x^* + d) - p(x^*)}.$$

If  $p$  is unbounded, the denominator tends to infinity while  $p'(x^* + d) < \gamma$ , so  $R(d) \rightarrow 0$ . If instead  $p'(x) > 0$  and  $p'(x) \rightarrow 0$ , then for  $d > 1$  the denominator is bounded below by  $p(x^* + 1) - p(x^*) > 0$ , while the numerator tends to zero; again  $R(d) \rightarrow 0$ .

For completeness,  $R$  is nonincreasing. Put  $y = x^* + d$  and  $C = p(x^*)$ . Then

$$R'(d) = \frac{p''(y)[p(y) - C] - p'(y)^2}{[p(y) - C]^2} \leq 0,$$

because  $p'' \leq 0$ ,  $p(y) - C \geq 0$ , and  $p'(y)^2 \geq 0$ . Continuity therefore gives a threshold  $\bar{s} > s_m$  with  $\Delta_m > 0$  iff  $s_{m+1} > \bar{s}$ , allowing for the boundary equality at the threshold.

### A.2 The ratio inequality used for FOSD

Let  $x_i \geq y_i \geq 0$  and define  $X_m = \prod_{i=1}^m x_i$ ,  $Y_m = \prod_{i=1}^m y_i$ . The desired inequality is

$$\frac{\sum_{m=1}^k X_m}{\sum_{m=1}^N X_m} \leq \frac{\sum_{m=1}^k Y_m}{\sum_{m=1}^N Y_m}.$$

Cross-multiplication reduces it to

$$\sum_{m \leq k} \sum_{n > k} (X_n Y_m - X_m Y_n) \geq 0.$$

For  $n > m$ ,

$$X_n Y_m - X_m Y_n = X_m Y_m \left( \prod_{j=m+1}^n x_j - \prod_{j=m+1}^n y_j \right) \geq 0.$$

This is the shortest self-contained proof of the exchange-of-summation lemma used in Appendix A.

### A.3 Exogenous-transition benchmark

If upward rates are fixed numbers  $\lambda_k$  rather than functions of effort, then

$$\pi_k = \pi_1 \prod_{i=1}^{k-1} \frac{\lambda_i}{\mu}$$

does not depend on AI assistance. The deterministic optimizer implies every  $p^*(s_k, a)$  is weakly increasing, hence their fixed weighted average is weakly increasing. Similarly,

$$\mathcal{E}(a) = \sum_k \pi_k (x^* - s_k - a)_+$$

decreases and eventually reaches zero. These conclusions isolate the endogenous transition channel.

## A.4 Relaxed IARA and boundary cases

For capped functions,  $p' = p'' = 0$  beyond a saturation point, so  $A = -p''/p'$  is formally  $0/0$ . The paper's relaxed convention assigns infinite ARA to the flat region and permits weak monotonicity of  $A$ . In the interior effort case define  $D = qp''(x_1) + (1 - q)p''(x_0)$ .

- If  $D \neq 0$ , the exact derivative (4) remains valid.
- If  $D = 0$ , concavity forces both second derivatives to be zero locally. Under the relaxed shape restriction, either the good state is flat and the bad state is locally linear, yielding  $e_a^* = 0$  and  $p_a^* = 0$ , or both are on the same positive-slope linear segment, yielding  $e_a^* = -1$  and nonpositive productivity derivative.
- At a zero-effort corner,  $p_a^* = qp'(s + \bar{a}) \geq 0$  as before.

Thus relaxed IARA gives weakly decreasing then increasing productivity; relaxed DARA gives weakly increasing productivity.

## A.5 Two partially IARA production functions

For the translog form

$$p(x) = \log x + b(\log x)^2, \text{quad} x > 0, b > 0,$$

write  $z = 1 + 2b \log x$ . Then

$$p'(x) = \frac{z}{x}, \quad p''(x) = \frac{2b - z}{x^2}, \quad \text{quad} A(x) = \frac{z - 2b}{xz}.$$

Differentiation gives

$$A'(x) = \frac{-z^2 + 2bz + 4b^2}{x^2 z^2}.$$

The numerator is positive when  $z < (1 + \sqrt{5})b$  and concavity requires  $z > 2b$ . This yields the paper's partial IARA region, together with the positivity conditions on  $p$  and  $p'$ .

For the transcendental multiproduct form

$$p(x) = e^{b_0 + b_1 x + b_2 x^2},$$

one can compute  $p' = p(b_1 + 2b_2 x)$  and  $p'' = p[(b_1 + 2b_2 x)^2 + 2b_2]$ . Substitution into  $A = -p''/p'$  and differentiation identify parameter and domain regions with  $p' > 0$ ,  $p'' < 0$ , and  $A' > 0$ . The lesson is local: globally standard production functions can satisfy IARA on economically relevant operating ranges.

## A.6 Why the capped-linear example can be smoothed

Let  $p(x) = \min\{1, \beta x\}$  and choose a small  $\delta > 0$ . Replace the kink on  $[1/\beta - \delta, 1/\beta + \delta]$  by a concave differentiable bridge  $\tilde{p}$  that differs from  $p$  by at most  $\delta$  and has weakly increasing ARA under the capped convention. At low proficiency  $a_\ell = 0$ , the optimizer stays within the bridge and productivity is at least  $1 - \delta$ . At high proficiency  $a_h = 1/\beta$ , if  $q > (\beta - \gamma)/\beta$ , the optimizer is within  $[0, \delta]$  and productivity is at most

$$q + (1 - q)\beta\delta.$$

Therefore

$$\frac{p^*(0, a_h, q)}{p^*(0, a_\ell, q)} \leq \frac{q + (1 - q)\beta\delta}{1 - \delta}.$$

First choose  $q < \varepsilon/2$  but above the reliability cutoff, then choose  $\delta < \varepsilon/[2(\beta + \varepsilon)]$ . The ratio is below  $\varepsilon$ . This shows the paradox is not an artifact of the exact kink.

### A.7 Monotonicity of ex-ante effort in $s$ , $\bar{a}$ , and $q$

For any posterior reliability  $r$ , define

$$\Omega(s, \bar{a}, r) = \{e \geq 0 : rp'(s + e + \bar{a}) + (1 - r)p'(s + e) \geq \gamma\}.$$

Concavity makes this set an interval beginning at zero, and the largest optimizer is its supremum. Increasing  $s$  or  $\bar{a}$  lowers the marginal-return expression pointwise. Increasing  $r$  also lowers it because  $p'(s + e + \bar{a}) \leq p'(s + e)$ . Hence  $\Omega$  shrinks in each argument and  $e^*(s, \bar{a}, r)$  is weakly decreasing in  $s$ ,  $\bar{a}$ , and  $r$ .

### A.8 Constant verification: the full monotonicity argument

When  $v(s) \equiv v \geq 1/2$ ,  $q_1$  and  $q_0$  are independent of  $s$  and  $q_1 \geq q_0$ . Thus

$$e_b^*(s) = \rho_1 e^*(s, \bar{a}, q_1) + \rho_0 e^*(s, \bar{a}, q_0)$$

is a fixed convex combination of functions decreasing in  $s$ . It is therefore decreasing. Moreover,  $q_1$  and  $q_0$  increase with the prior  $q$ , each conditional effort falls with its reliability, and the weight on the smaller effort  $e^*(\cdot, q_1)$  rises. Hence expected effort also falls with  $q$ . Only the first monotonicity is needed to prove unimodality.

## B Notation and assumption index

| Symbol                     | Meaning                                                                   |
|----------------------------|---------------------------------------------------------------------------|
| $s, s_k$                   | current skill and the $k$ th ordered skill state                          |
| $e, e^*$                   | effort and a largest utility-maximizing effort                            |
| $a, \bar{a}$               | deterministic assistance; proficiency in the good realization             |
| $q$                        | prior reliability, $\mathbb{P}(a = \bar{a}) = q$                          |
| $p(x)$                     | concave production function of total input                                |
| $\gamma$                   | constant marginal cost of effort                                          |
| $x^*$                      | largest maximizer of $p(x) - \gamma x$                                    |
| $\lambda(e)$               | upward skill-transition rate                                              |
| $\mu$                      | downward skill-decay rate                                                 |
| $\pi_k$                    | stationary probability of skill state $s_k$                               |
| $\mathcal{E}, \mathcal{P}$ | stationary effort and productivity                                        |
| $I_m$                      | AI-assistance interval with exactly the lowest $m$ states exerting effort |
| $\Delta_m$                 | sensitivity gap between skill development and productivity                |
| $A(x)$                     | absolute risk aversion, $-p''(x)/p'(x)$                                   |
| $v(s)$                     | skill-dependent verification accuracy                                     |
| $q_1, q_0$                 | posterior reliabilities after positive and zero signals                   |
| $e_b^*$                    | expected effort of a Bayesian signal follower                             |
| $\omega$                   | marginal return gap, $(\beta - \gamma) - q\beta$                          |
| $\tilde{s}$                | skill at which one posterior crosses the reliability cutoff               |



## C Research scope and adjudication

These notes use the authors’ v1 preprint and its TeX source as the sole substantive source. The main text and supplementary Appendices A–C were checked in full. Appendix D, “Extension to convex cost functions,” and its numerics were excluded by request. The optimizer, stationary-distribution formulas, FOSD result, key derivatives, two-state threshold, posterior crossings, and mode arguments were independently re-derived. Empirical papers cited in the introduction of the source paper were not independently audited; no empirical claim is needed for the mathematical conclusions in this tutorial.

What is well established within the model: the deterministic optimizer; the stationary birth–death formula; the derivative identity governed by ARA; and the adjacent-ratio characterization of modes. What remains model-dependent: additive substitutability, myopia, the binary unreliability structure, and the capped-linear functional form used for the polarization theorem. The lecture plan, dependency map, diagnostic checks, and exercises are pedagogical additions.

## References

- [1] Ali Aouad, Thodoris Lykouris, and Huiying Zhong. *Human–AI Productivity Paradoxes: Modeling the Interplay of Skill, Effort, and AI Assistance*. arXiv:2605.11350v1, 12 May 2026.
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